K-Nearest Neighbors (KNN) — MCQs

1. KNN is primarily considered a:

A. Supervised learning algorithm
B. Unsupervised learning algorithm
C. Reinforcement learning algorithm
D. Generative learning algorithm

2. What does the value K represent in KNN?

A. Number of features
B. Number of classes
C. Number of nearest neighbors considered
D. Number of training iterations

3. KNN is commonly referred to as a:

A. Parametric algorithm
B. Lazy learning algorithm
C. Generative algorithm
D. Gradient-based algorithm

4. Why is KNN called a lazy learner?

A. It does not require training data
B. It postpones most computation until prediction time
C. It automatically reduces the dataset size
D. It does not calculate distances

5. Which distance measure is most commonly used with KNN for continuous numerical features?

A. Euclidean distance
B. Cosine similarity only
C. Hamming distance only
D. Jaccard distance only

6. Suppose $K = 5$ in a KNN classification problem. What does the algorithm do for a new sample?

A. Considers all training samples
B. Considers the 5 nearest training samples
C. Randomly selects 5 training samples
D. Selects the 5 farthest samples

7. In KNN classification, the predicted class is generally determined by:

A. Mean of all feature values
B. Majority vote among the nearest neighbors
C. Random selection
D. The class with the highest numerical label

8. Consider $K = 3$. The three nearest neighbors of a test point belong to classes:

$$A,\ A,\ B$$

What class will standard KNN predict?

A. A
B. B
C. Cannot be determined
D. Both A and B

9. What is a major reason for normalizing or standardizing features before applying KNN?

A. KNN cannot process numerical values
B. Features with larger scales can dominate distance calculations
C. It reduces the value of *K*
D. It converts KNN into a parametric model

10. If $K = 1$, a KNN classifier:

A. Uses the entire training dataset
B. Uses only the nearest training sample
C. Uses the two nearest samples
D. Randomly selects one sample

11. What is likely to happen if *K* is chosen to be extremely small, such as $K = 1$?

A. The model may become sensitive to noise and outliers
B. The model always generalizes better
C. The model becomes completely independent of the training data
D. The model cannot classify new samples

12. What is likely to happen if *K* is chosen to be very large?

A. The model becomes more sensitive to individual points
B. The model may become overly influenced by the overall class distribution
C. Training error necessarily becomes zero
D. Distance calculations are no longer required

13. Which statement about KNN training is MOST accurate?

A. KNN learns a fixed set of model weights during training
B. KNN generally stores the training data and performs computation during prediction
C. KNN requires gradient descent
D. KNN learns cluster centroids

14. Consider two features:

- Age: 20–80
- Annual income: 20,000–2,000,000

When using Euclidean distance in KNN without scaling, what is the likely issue?

A. Age will dominate the distance
B. Income may dominate the distance
C. Both features will automatically contribute equally
D. KNN will ignore both features

15. A KNN classifier achieves 100% training accuracy but poor test accuracy. Which is the MOST likely explanation?

A. Underfitting
B. Overfitting
C. The value of *K*is necessarily too large
D. KNN cannot be used for classification

16. Which change would generally make a KNN model more sensitive to local variations in the training data?

A. Increasing *K*substantially
B. Decreasing *K*
C. Removing the distance calculation
D. Increasing the number of classes

17. A new data point is equally distant from two neighbors belonging to different classes. What issue can arise?

A. Distance cannot be calculated
B. A tie may occur during voting
C. KNN automatically changes *K*
D. The data point is always rejected

18. Which statement about KNN and dimensionality is TRUE?

A. KNN always performs better as the number of dimensions increases
B. KNN can suffer from the curse of dimensionality
C. KNN is unaffected by irrelevant features
D. KNN automatically removes irrelevant features

19. Suppose the five nearest neighbors of a test sample have labels:

$$A,\ B,\ B,\ A,\ B$$

If $K = 5$, what will standard majority-vote KNN predict?

A. A
B. B
C. Both A and B
D. Cannot be determined

20. Tricky: Suppose $K = 3$gives 95% training accuracy and 80% test accuracy, while $K = 15$ gives 88% training accuracy and 90% test accuracy. Which statement is MOST reasonable?

A. $K = 3$is necessarily better because its training accuracy is higher
B. $K = 15$**may generalize better to unseen data**
C. $K = 3$must be selected because smaller $K$is always better
D. $K = 15$is overfitting because its training accuracy is lower